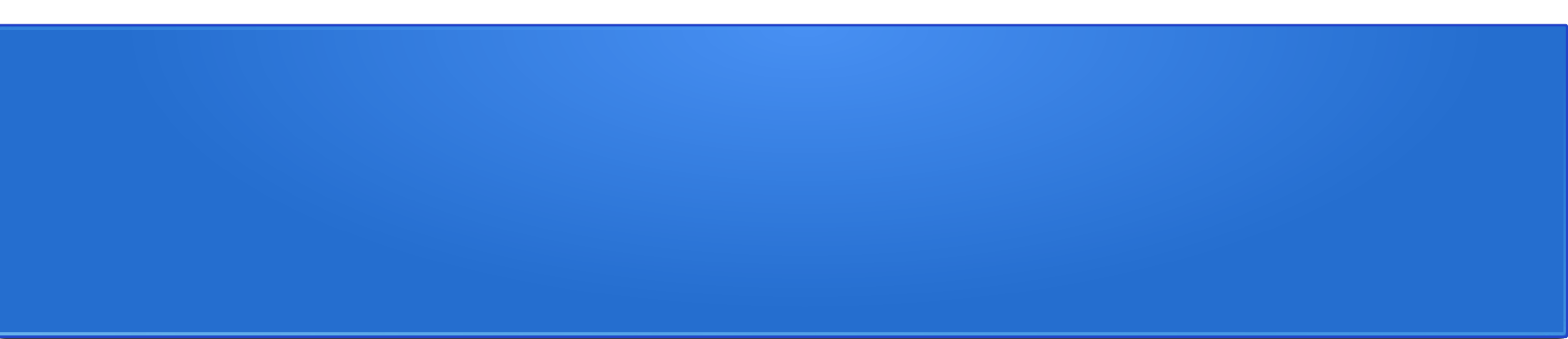
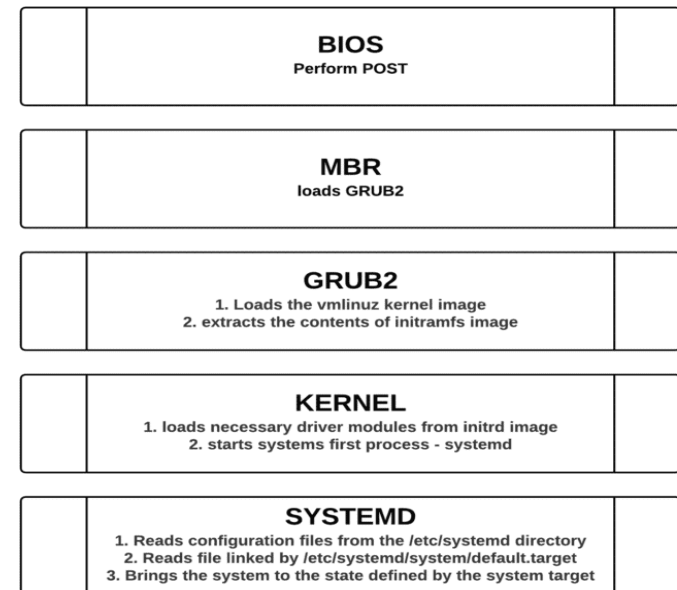




Boot Sequence

Introduction

- It is important to understand the Linux boot process to troubleshoot boot problems.



BIOS

- The BIOS (Basic Input/Output System)
 - performs the POST (power on self test) to detect, test and initialize system hardware components.
 - Loads the MBR (Master boot record).

Master boot record (MBR)

- Master Boot Record (MBR) is the first 512 bytes of the boot drive that is read into memory by the BIOS.
- The first 64 bytes contain the partition table for the disk.
- The last two bytes are the “Magic Number” which is used for error detection.
- MBR discovers the bootable device and loads the GRUB2 boot loader into memory and transfers control over to it.



GRUB2 Bootloader

- The default bootloader program used on RHEL 7 is GRUB 2.
- GRUB stands for GRand Unified Bootloader.
- The GRUB 2 configuration file is located at `/boot/grub2/grub.cfg` (Do not edit this file directly).
- GRUB 2 menu-configuration settings are taken from `/etc/default/grub` when generating `grub.cfg`.

GRUB2 Bootloader

Cont.

- Sample /etc/default/grub file :

```
GRUB_TIMEOUT=5
```

```
GRUB_DISTRIBUTOR="$(sed 's, release .*$,,g' /etc/system-release)"
```

```
GRUB_DEFAULT=saved
```

```
GRUB_DISABLE_SUBMENU=true
```

```
GRUB_TERMINAL_OUTPUT="console"
```

```
GRUB_CMDLINE_LINUX="crashkernel=auto rhgb quiet"
```

```
GRUB_DISABLE_RECOVERY="true"
```

- If changes are made to any of these parameters, you need to run `grub2-mkconfig` to re-generate the `/boot/grub2/grub.cfg` file.

```
# grub2-mkconfig -o /boot/grub2/grub.cfg
```

GRUB2 Bootloader

Cont.

- GRUB2 searches the compressed kernel image file also called as vmlinuz in the /boot directory.
- GRUB2 loads the vmlinuz kernel image file into memory and extracts the contents of the initramfs image file into a temporary, memory-based file system (tmpfs).
- The initial RAM disk (initrd) is an initial root file system that is mounted before the real root file system.

GRUB2 Bootloader Cont.

- The job of the initial RAM file system is to preload the block device modules, such as for IDE, SCSI, or RAID, so that the root file system, on which those modules normally reside, can then be accessed and mounted.
- The initramfs is bound to the kernel and the kernel mounts this initramfs as part of a two-stage boot process.
- The dracut utility creates initramfs whenever a new kernel is installed.
- Use the `lsinitrd` command to view the contents of the image created by dracut

Kernel

- The kernel starts the `systemd` process with a process ID of 1 (PID 1)

systemd

- `systemd` is the parent of all processes on a system.
- It reads the file linked by `/etc/systemd/system/default.target` (for example, `/usr/lib/systemd/system/multi-user.target`) to determine the default system target (equivalent to run level).
- The system target file defines the services that `systemd` starts.

Systemd cont

- `systemd` brings the system to the state defined by the system target, performing system initialization tasks such as:

- Setting the host name
- Initializing the network
- Initializing SELinux based on its configuration
- Printing a welcome banner
- Initializing the system hardware based on kernel boot arguments
- Mounting the file systems, including virtual file systems such as the `/proc` file system
- Cleaning up directories in `/var`
- Starting swapping

systemd target units

- Target units have a .target extension.
- Target units allow you to start a system with only the services that are required for a specific purpose.
- To list the predefined systemd run level target units :

- `find / -name "runlevel*.target"`

`/usr/lib/systemd/system/runlevel4.target`

`/usr/lib/systemd/system/runlevel5.target`

`/usr/lib/systemd/system/runlevel6.target`

`/usr/lib/systemd/system/runlevel1.target`

`/usr/lib/systemd/system/runlevel2.target`

`/usr/lib/systemd/system/runlevel3.target`

`/usr/lib/systemd/system/runlevel0.target`

systemd target units

cont.

Target Units	Description
runlevel0.target, poweroff.target	Shut down and power off
runlevel1.target, rescue.target	Set up a rescue shell
runlevel[234].target, multi-user.target	Set up a nongraphical multi-user shell
runlevel5.target, graphical.target	Set up a graphical multi-user shell
runlevel6.target, reboot.target	Shut down and reboot the system

systemd target units cont.

- To view which target unit is used by default use :

```
# systemctl get-default  
multi-user.target
```

Notice that the default.target symbolic file points to graphical.target file.

```
# ls -l /etc/systemd/system/default.target
```

```
lrwxrwxrwx. 1 root root 40 Oct 11 02:02 /etc/systemd/system/default.target -> /usr/lib/systemd/system/graphical.target
```

systemd target units cont.

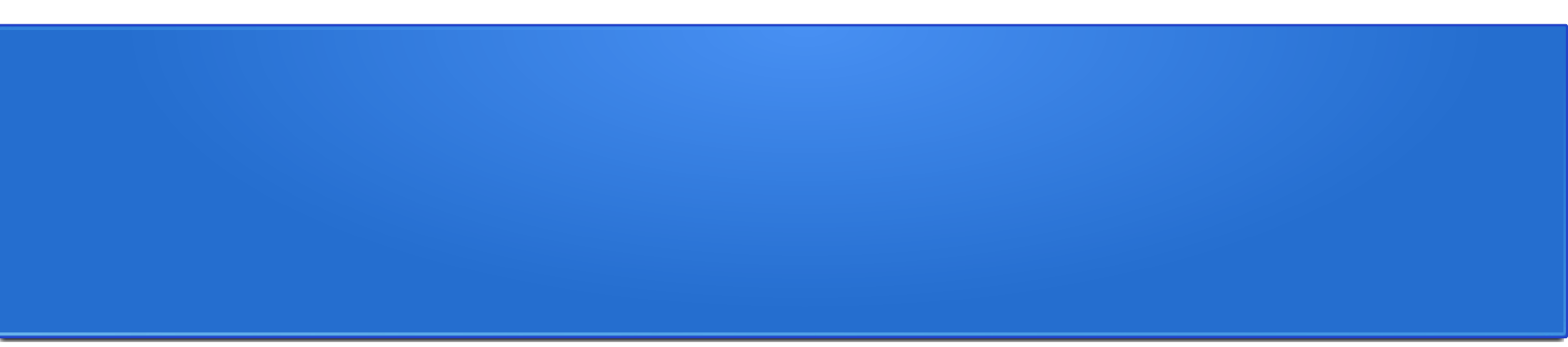
- To change the default target unit to the graphical.target unit

```
# systemctl set-default graphical.target
```

```
rm '/etc/systemd/system/default.target'
```

```
ln -s '/usr/lib/systemd/system/graphical.target' '/etc/systemd/system/default.target'
```

- To change currently active target
- `# systemctl isolate multi-user.target`



Services

To Start and Stop Services

- Start a service

- `# systemctl start [service_name]`

- Example

```
# systemctl start network
```

- Stop a service:

```
# systemctl stop [service_name]
```

- Example:

```
# systemctl stop network
```

To Check status of a service

```
# systemctl status  
[service_name]
```

- Example

-

- ```
systemctl status network
```

# Enabling or disabling auto-start of a service

- Enable auto-start a service at boot):

```
systemctl enable [service_name]
```

- Example:

```
systemctl enable network
```

- Disable auto-start a service at boot:

```
systemctl disable [service_name]
```

- Example:

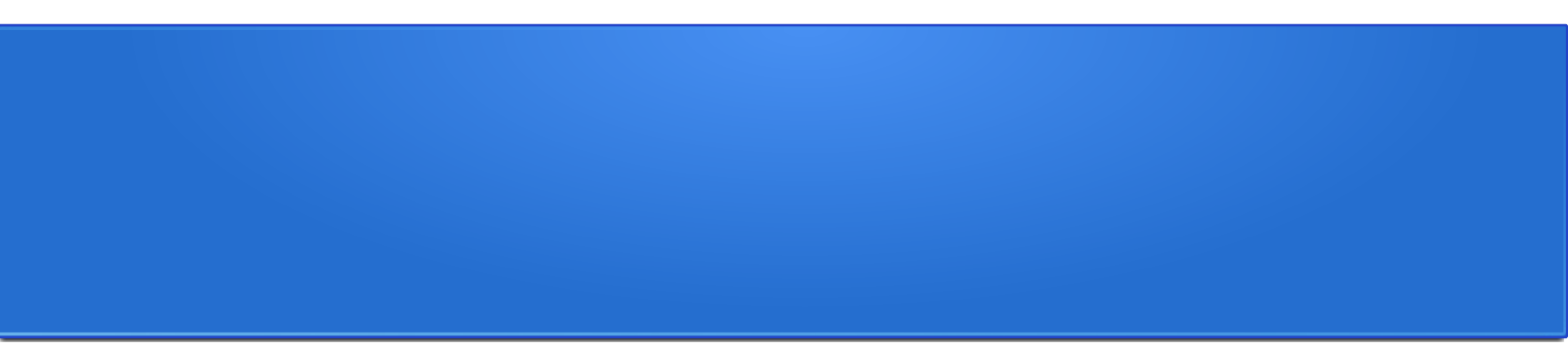
```
systemctl disable network
```

# Listing services

- List enabled/disabled status of all installed service units:

- 

```
systemctl list-unit-files --type service
```



# **Scheduling and Processes**

# Task automation

- Red Hat Enterprise Linux provides two utilities to manipulate and schedule tasks
  - The `at` service
    - Executes a task at a specific time
    - Uses the `at` command to set up the execution of a task
    - Is administered by the `atd` daemon
  - The `cron` service
    - Executes tasks at scheduled intervals.
    - Uses the `crontab` command to set up the execution of a task.
    - Is administered by the `crond` daemon

# Access to the `at` command

- `/etc/at.allow` and `/etc/at.deny` files restrict or allow users to run the `at` command.
- If `/etc/at.allow` exists and your username appears in it, you may use `at` command.
- If `/etc/at.allow` does not exist and `/etc/at.deny` does, then you must not be listed in `/etc/at.deny` to use `at`.
- If neither file exists, all users are denied to schedule jobs with `at`.
- The default installation supplies an empty `at.deny` file, thereby allowing the use of `at` to all users.

# Use `cron` for task automation

- Crontab files are stored in `/var/spool/cron`
- Avoid modifying these files directly, use `crontab -e` command

## ■ The cron file format

Min(0-59) Hours(0-23) Day(1-31) Month(1-12) Day(0-6) command

## ■ Example

```
30 6 * * 0 /home/rania/mys1
```



# The /etc/crontab File

- It automates the execution of system tasks. These tasks are often vital for the integrity of the operating system.

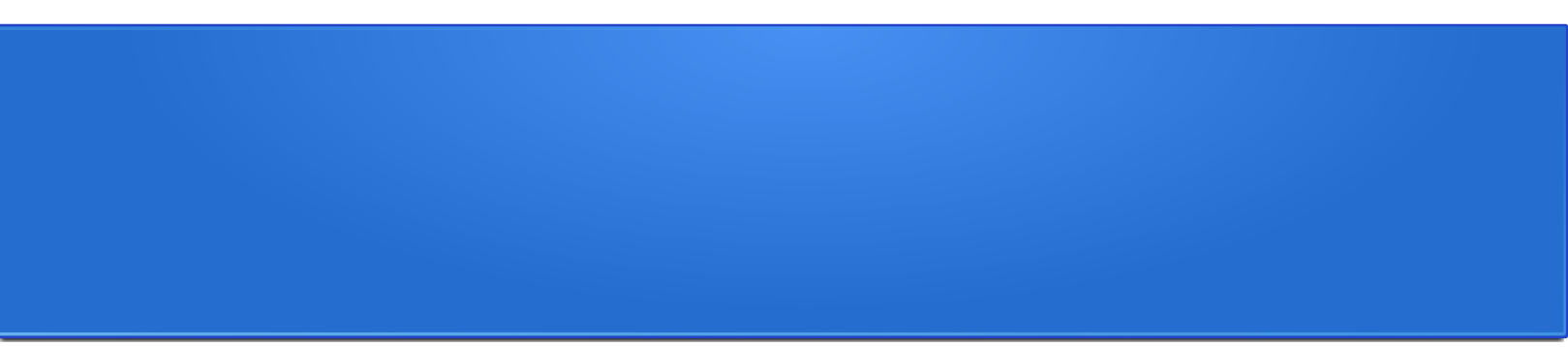
```
cat /etc/anacrontab
SHELL=/bin/bash
PATH=/sbin:/bin:/usr/bin:/usr/sbin
MAILTO=root
HOME=
```

```
•#period in days delay in minutes job-identifier command

1 5 cron.daily nice run-
parts /etc/cron.daily

7 25 cron.weekly nice run-parts
/etc/cron.weekly

@monthly 45 cron.monthly nice run-parts
/etc/cron.monthly
```

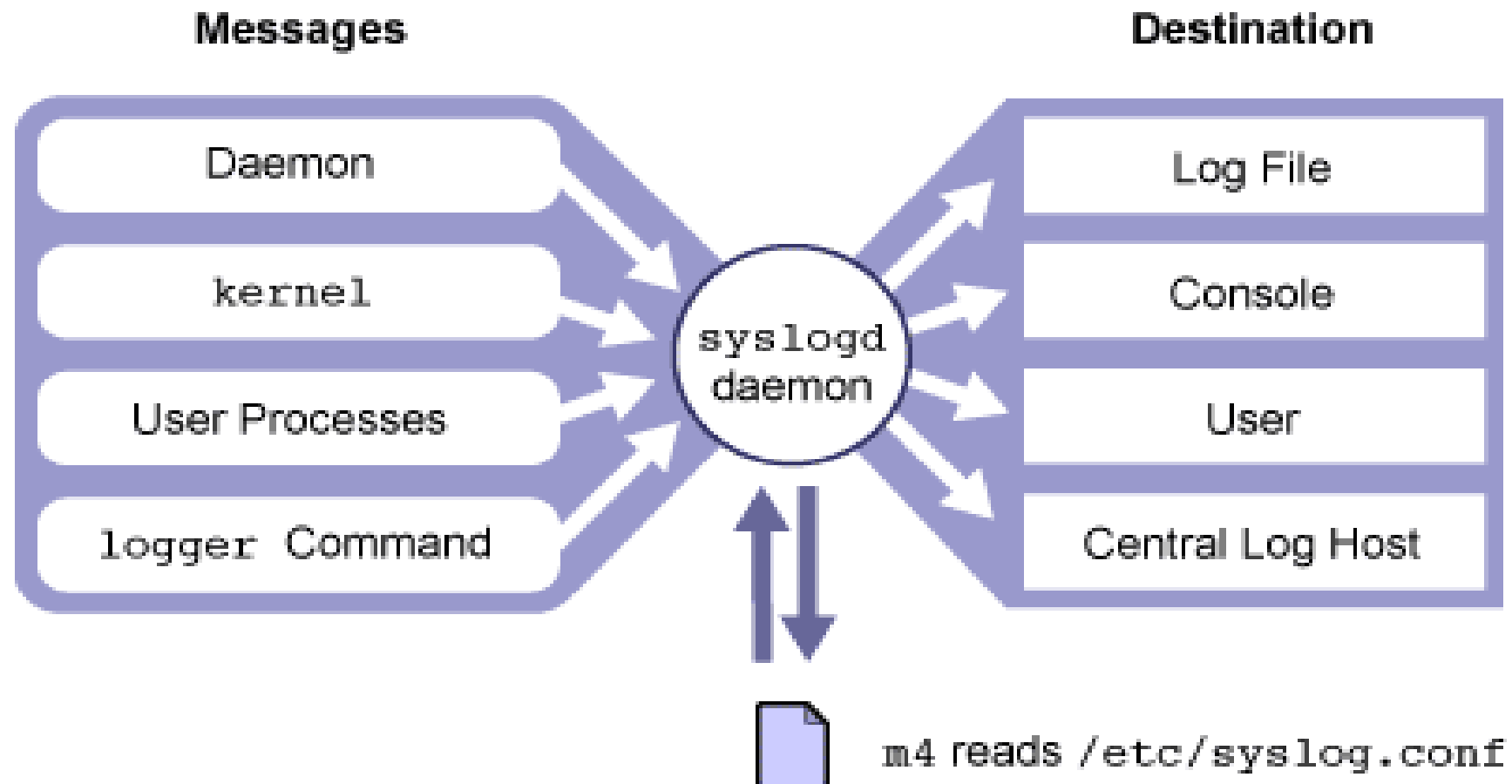


# **Managing System Logs**

# Introduction

- The `rsyslog` system messaging features track system activities and events.
- Log messages can be manually generated using `logger` command

# How rsyslogd Works ?



# rsyslogd Configuration File

- `/etc/rsyslog.conf` file
- Selector
  - Facility: category of system processes that generate the messages
  - Example: `kern, mail, daemon, cron, local0-7, *, ...`
  - Level: severity or importance of the message
  - Example: `emerg, alert, crit, err, warning, ...`
- Action: where to send a message
  - Example: `/pathname, @hostname, username, *`
  - `/pathname` must already exist

# Monitoring a rsyslog File in Real Time

- You can monitor the designated `rsyslog` file in real time using the command `tail -f`.
- The `tail -f` command holds the file open so that you can view messages being written to the file by the `syslogd` daemon.

```
tail -f /var/adm/messages
```

```
jun 14 13:15:39 host1 inetd[2359]:[ID 317013 daemon.notice] telnet [2361] from 192.9.200.1 45800
Time/Date localhostname Pname[pid] MSG ID facility.level incoming request [ppid] IP Port name
```

## Adding One-Line Entries to a System Log File

- The `logger` command enables you to send messages to the `rsyslogd` daemon.
- A system administrator can write administrative shell scripts that report the status of backups, or other functions by using the `logger` command.

# Adding One-Line Entries to a System Log File Cont'd

```
logger [-i] [-f file] [-p priority] [-t tag] [message]
```

| Option      | Description                                                                |
|-------------|----------------------------------------------------------------------------|
| -i          | Logs the process Id of the logger command with each line                   |
| -f filename | Uses the content of the file as the message body                           |
| -p priority | Enters the message with the specified priority, the default is user.notice |



# Identify system log files

- The `/var/log/dmesg` file

- Kernel log messages.
- This log file is written after system boot.
- It contains messages from the kernel that were generated during the boot process.

- The `/var/log/messages` file

- It the standard system log file.
- It contains messages from most of your system's services.

# Identify system log files cont'd

- The `/var/log/maillog` file

- Mail system messages
- It contains messages and errors from your mail service

- The `/var/log/secure` file

- It contains messages and errors from security-related services such as login, network connections (`xinetd`)
- Very useful in detecting and investigating attempts to gain unauthorized access to the system.

# Maintaining logs

- Left unchecked, system logs will grow until you run out of disk space.
- Red Hat Enterprise Linux provides `logrotate`, a powerful tool for log file maintenance.
- Log files are rotated at predefined intervals.
- `logrotate` can be configured to compress old log files to conserve filesystem space.

# Maintaining logs cont'd

## ■ Example

- /var/log/messages is rotated weekly to /var/log/messages.1, with older log files rotated to /var/log/messages.2, etc.
- Generally, four weeks of log files are kept; older rotated files are deleted.
- When you install a new system service by Red Hat Package Manager (RPM) that generates its own log file, it should "pre-configure" itself automatically for log file rotation by placing a configuration file in the `/etc/logrotate.d` directory.

# A sample of logrotate.conf File

```
more /etc/logrotate.conf
see "man logrotate" for details
rotate log files weekly
weekly

keep 4 weeks worth of backlogs
rotate 4

create new (empty) log files after rotating old ones
create

uncomment this if you want your log files compressed
#compress

RPM packages drop log rotation information into this directory
include /etc/logrotate.d

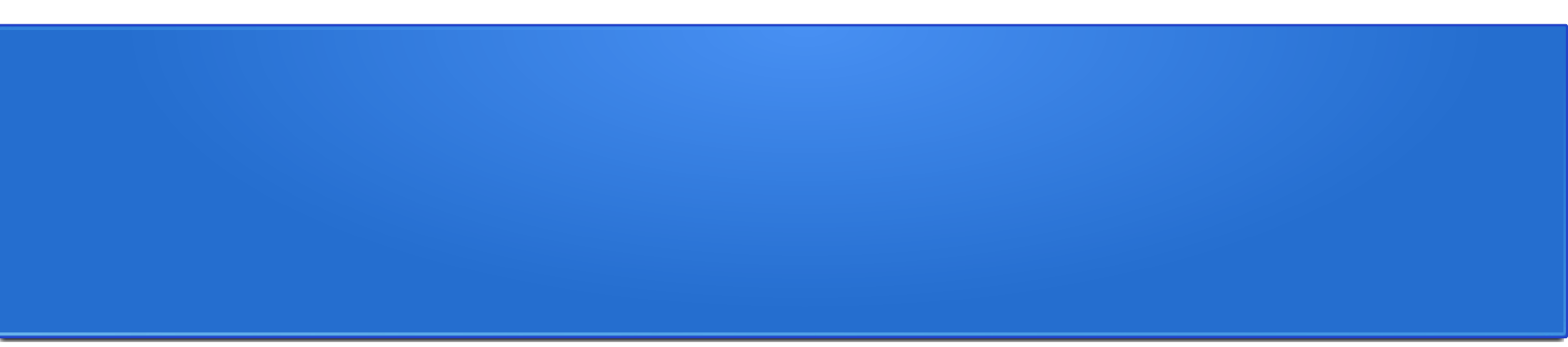
no packages own wtmp -- we'll rotate them here
/var/log/wtmp {
 monthly
 create 0664 root utmp
 rotate 1
}
```

# Maintaining Logs

- Monitoring system logs is an onerous but important task.
- If you do not properly monitor your logs, you might miss
  - Security problems
  - Hardware problems
  - Software problems

# Maintaining Logs cont'd

- The logwatch utility, can be installed, parses log files and generates a report that can either be printed or sent to a user via e-mail.
  - By default, logwatch runs nightly as a result of the 00-logwatch script located in the `/etc/cron.daily` directory. The result of this nightly execution is a report e-mailed to the root user.
  - The logwatch utility can be configured to detect almost any type of activity with the `/etc/logwatch/conf/logwatch.conf` file.
- \* See `/usr/share/doc/logwatch-version` (where *version* is the version of logwatch that you are using) for information on writing log filters.



# Network Configuration



# Network Interfaces

## ■ Interface names

- eth0
- eth1
- eth2
- Wlan0 (the first wireless device)
- virbr0 (internal bridge set up for virtual hosts)
- ...

## ■ To view MAC address

- Use `ifconfig` command
- Use `/sbin/ip` command

# Network Interfaces Commands

- /sbin/ip is used to show or temporarily modify devices

```
ip addr show eth0
```

- The `ifconfig` command displays and configures IP addresses on network interfaces.

- To display the network settings of all active network devices

```
ifconfig
```

- To see both active and inactive network device setting

```
ifconfig -a
```

- To bring up or down a network interface

```
ifdown eth0
```

```
ifup eth0
```

# Hostname Resolution

- The hostname command displays or temporarily modifies hostname

```
hostname
```

```
iti.gov.eg
```

```
cat /etc/hosts
```

```
192.168.0.250 iti.gov.eg iti #Added by
NetworkManager
```

```
127.0.0.1 localhost.localdomain
localhost
```

```
.....
```

# Modifying Network Configuration

- NetworkManger provides network status information and graphical configuration tools that can create, edit and remove connections and interfaces.
- To change from using DHCP to static IP address
  - Right-click the NetworkManger icon in the top Panel and select **Edit connections**
  - On the **wired** tab, select **System eth0** and click **Edit** tab
  - Select **IPv4 Setting** tab
  - On the **Method** drop-down menu, change **Automatic (DHCP)** to **Manual**
  - Under **Addresses** click **Add** and enter the Ipv4 address, netmask gateway router and DNS server to use
  - Make sure that **Connect automatically** is checked so the interface starts at boot and **Available to all users** is checked so that it is available system-wide
  - Click **Apply** to apply your changes

# Modifying Network Configuration

## cont'd

- Editing interface configuration files `/etc/sysconfig/network-scripts/ifcfg-<interface-name>`

Any

DHCP

Static

```
DEVICE=eth0
ONBOOT=yes
HWADDR=52:54:00:0A:00:0A
MODE=Managed
```

```
BOOTPROTO=dhcp
```

```
BOOTPROTO=static
IPADDR=192.168.0.250
PREFIX=24
GATEWAY=192.168.0.254
DNS1=192.168.0.254
```

# Modifying Network Configuration cont'd

- The `/etc/sysconfig/network` is used to specify hostname and may specify a static default gateway

```
cat /etc/sysconfig/network
```

```
NETWORKING=yes
HOSTNAME=iti.gov.eg
GATEWAY=192.168.0.254
```

- if an entry is not found in `/etc/hosts`, then the DNS will be responsible for associating hostnames with IP addresses

```
more /etc/resolv.conf
```

```
domain gov.eg
```

```
search gov.eg
```

```
nameserver 213.131.65.20
```

```
nameserver 163.121.12.2
```

- When changing any system configuration, remember to:
  - Modify the configuration file
  - Restart a service
  - Verify the change

# Network Troubleshooting

## ■ IP Address and Subnet mask

- Test using ping command
- Check ip addr
- Fix by editing in /etc/sysconfig/network-scripts/ifcfg-\*

## ■ Routing/Default Gateway

- Test using traceroute command
- Check ip route
- Fix by editing in /etc/sysconfig/network-scripts/ifcfg-\*

## ■ Name Resolution

- Test using host command
- Check /etc/hosts and /etc/resolv.conf
- Fix by editing in /etc/sysconfig/network-scripts/ifcfg-\*

# Useful Commands

```
ping -c 2 192.168.0.250
```

```
... .
```

```
traceroute -Tn www.redhat.com
```

```
... .
```

```
#host ns1.redhat.com
```

```
ns1.redhat.com has address 66.187.233.210
```

```
getent hosts ns1.redhat.com
```

```
nslookup ns1.redhat.com
```

```
Server: 216.99.225.30
Address: 216.99.255.30#53
```

```
Non-authoritative answer:
Name: ns1.redhat.com
Address: 66.187.233.210
```



# Special Permissions

- SUID on an executable file
- SGID on an executable file
- SGID on a directory
- Sticky bit on a directory
- Example

```
chmod u+s executable1
ls -l
-rwsr-xr-x ...
chmod g+s executable1
ls -l
-rwxr-sr-x ...
chmod g+s directory1
ls -l
drwxr-sr-x ...
chmod o+t directory1
ls -l
drwxrwxrwt ...
```

# Access Control Lists

- Use ACLs to generate customized permission sets
- To enable ACLs on a filesystem, the filesystem must be mounted with the `acl` option.

## ■ Example

```
mount -o remount,acl /home
getfacl /home/sherine/fname
...
u::rwx # applies to owner
u:ahmed:rw- # applies to user ahmed
u:3142:--- # applies to user id 3142

g::rw- applies to file group
g:telecom:rwx applies to telecom group
g:10:rw- applies to group id 10

O:rwx applies to everyone else
```

# Access Control Lists Cont'd

```
setfacl -m u:user1:rw file1 # grants rw to user1

setfacl -m g:group1:rx directory1 # grants rx to group group1

setfacl -x u:user1 file1 # remove user1 acl entry
```

## ■ Permission Precedence

1. if owner, then the file's owner permission apply
2. else if user has an acl entry, then the user acl applies as long as it is permitted by **mask**
3. else if matches file's group or acl group entry then it applies
4. otherwise the file's other permissions apply.

# Access Control Lists Cont'd

## ■ The ACL mask

- Mask limits the maximum permissions that both the group that owns the file and that supplementary users and groups in ACLs can have.
- `getfacl` command displays the current mask `mask::perm`
  - \* **changing group permissions on a file with an ACL by using `chmod`, it actually changes the mask, which limits the max. permissions of all groups and supplementary users**

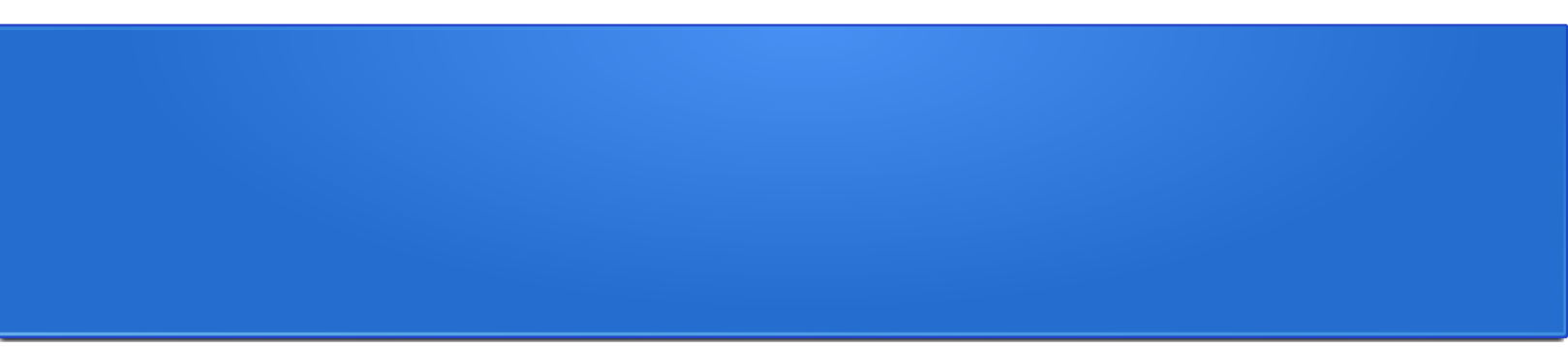
# Access Control Lists Cont'd

## ■ Default ACLs (Inheritance)

- A directory can have a default ACL entries that are automatically set on new files created in that directory

```
setfacl -m d:u:ahmed:rw directory
```

```
setfacl -m d:u:ahmed:rx directory
```



**Thanks :)**